EECS 388 Lab #5

Servo Motor Control

In this lab, we will control a small servo motor using PWM. For an RC car, a servo motor is typically used to control the car's steering.

Task 0: Download starter code and read Servo Motor Data Sheet

Task 0.0 Download starter code

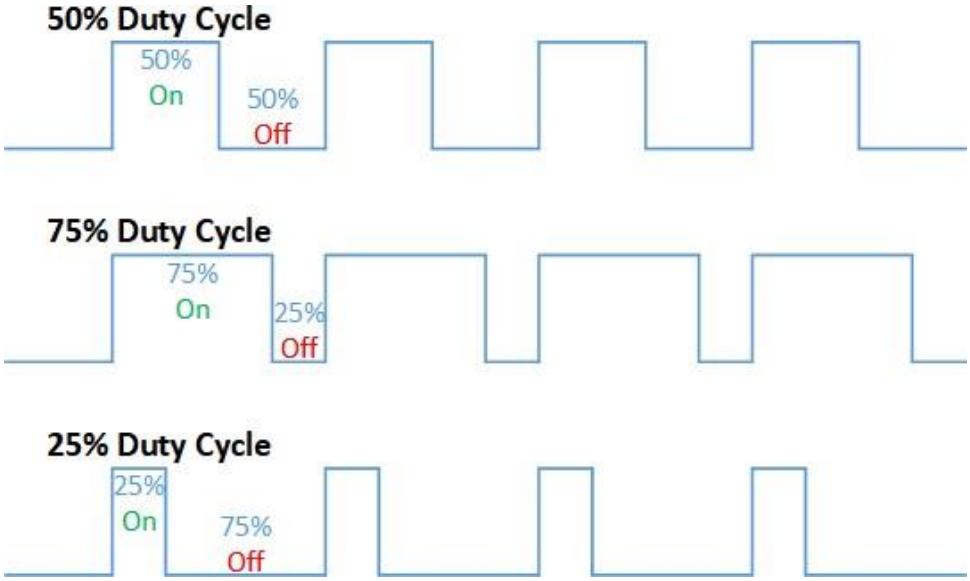
Log into Canvas and go to the EECS 388 course page. Go to "Files>Labs" and under "Lab 5" click on **lab05.zip** to download the lab source code. Extract the files into a folder on your PC and open the folder in **VSCode**, same as with Lab 2-4.

Task 0.1 Read through SG90 Servo Motor Data Sheet

The datasheet can be found in the **lab05/docs/sg90_microserver_datasheet.pdf.** Be sure to look at the wiring, PWM period, and duty cycle.

Pulse-width modulation (PWM) is a way to encode analog signals using digital outputs. Although a digital output can be either ON or OFF at a time, by altering the durations of ON and OFF we can encode an arbitrary analog signal.

In the diagram below assume the ON value is 5V. If the digital output is ON for 50% of the duration (top figure), its voltage is rated as 2.5V on average. If the duty cycle is 75%, its voltage is 3.75V on average. If the duty cycle is 25%, its voltage is 1.25V on average.



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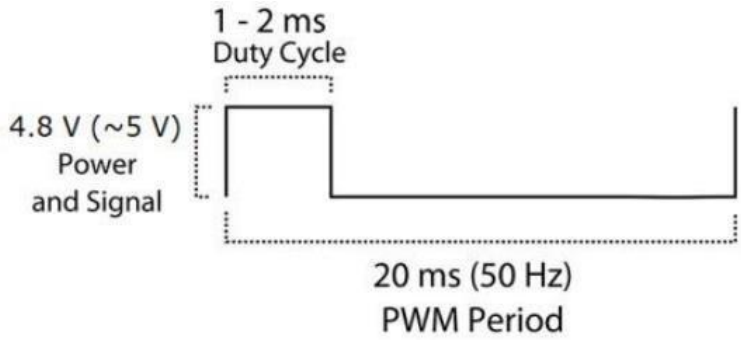
Note: the PWM wire may be yellow depending on the specific servomotor model used.

A servo motor receives a PWM signal whose value represents the desired angle of the servo motor. The PWM value ranges may vary depending on the specific servo, but most RC servo motors, including the one we use in this lab, operate between **544 us** (0 degrees) to **2400 us** (180 degree) duty cycle at a 20 ms period (50 Hz).



Task 1: Servo control programming

Task 1.0 Connect **the Servo Motor to your board**
Connect the servo motor's Vcc (red, center), ground (brown), and PWM (orange) to the Arduino board's 5V, GND, and pin6 lines, respectively.

<u>CAUTION</u>: Driving the servo motor with too high of a duty cycle can damage the motor. Review your code and wiring with your GTA before loading it!

Task 1.1 Program the servo() function
For this lab, we will manually toggle the GPIO6 on Pin 6 to create the PWM waveform. You will need to use provided library functions: **`gpio_write`**, **delay_us, and delay_ms**. The **`delay_us`** function is similar to **`delay_ms`**, except the input is in micro-seconds instead of miliseconds.

Note: The delay_us() is tuned specifically for the 16MHz Arduino UNO and has a limitation that it can only be used for values from 0 – 16,000 microseconds.

Using these functions, you should output the ON signal (5V) for a specific duty cycle duration, followed by the OFF signal for the remainder of the PWM period. This will create a PWM signal, which controls the servo motor's angle.

The project code you downloaded already includes the basic skeleton of the program that will sweep different angles, remaining at each angle for about a 1 second duration.

You will need to complete the **`servo()`** function in the **eecs_388_pwm.c** to complete the lab. For the demo, you need to show that after running your code, the servo motor rotates by fixed 30 degree increments going from 0 degrees to 180 degrees, and then cycles back to 0 degrees and repeats the rotations.

Task 1.2 Additional LED Colors – Optional Bonus – 1 additional point
Remove the servo motor before attempting this and comment out the servo() function call.

Create a separate led() function to adjust the brightness of the various LEDs to make different color combinations. You can create a loop to cycle through these new additional colors.

Try to get additional LED colors by using PWMs on the GPIOs at different duty cycles. The duty cycles may need adjustments.

- Blue (full) + Green (full) + RED (off) = Teal
- Red (full) + Green (10%) + Blue (Off) = Orange
- Red (30%) + Green (80%) + Blue (On) = Light Blue
- Red (80%) + Blue (20%) + Green(Off) = Hot Pink
- Red (100%) + Blue (100%) + Green (60%) = Purple

Submission – 30 points (Max 32.5/30 with bonus)

Attendance: To obtain the 25% (7.5 points), you must attend the entire lab period and work toward the demo. If you are more than 30 minutes late, it will be counted as absent.

Demo: To obtain the 50% (15 points) for this lab, you must show your GTA the working servo motor, rotating by 30 degrees from 0-180 degrees and back. You will also need to be able to answer some of the following questions:

- What is the maximum/minimum angle position of the servo specified in the code?
- What is the maximum/minimum possible duty cycle duration as specified in the code?
- How are you sending the servo output signal to the GPIO of the Arduino? (specify the lines of your code which are doing these)
- What is the purpose of using the delay_us() function in the code? Is this a blocking or non-blocking call?
- How did you calculate the required delay in microseconds that the GPIO needed to be ON?

Code: To obtain the 25% (7.5 points) go to the "Assignments" section on Canvas and submit your modified **eecs_388_pwm.c** file to the Lab 05 submission link. Also, make sure that you have shown your completed work to your respective GTA for a demo.

Screenshots/PDF/Text files will **NOT** be allowed/graded as lab code submission. You need to submit modified **eecs_388_pwm.c** file **only**.

Bonus:
To obtain the 5% (1.5 points) you need to submit the files to Canvas by the end of your lab period. You must demo your working code on the board to your GTA before submitting the code to canvas.

You can obtain an additional 1 point by implementing the bonus task of displaying different LED colors and demoing it to your GTA. To get this bonus point, you need to show at least 3 of the new colors listed in the bonus task.